Surface Energy of Normal-Superconducting Interfaces

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Ginzburg-Landau Theory for Superconductor

In this text, we study the difference between type-I and type-II superconductors by investigating the energy cost of planer interface forming between the superconducting and normal state of the material. We will start by looking at a superconductor placed in a neutral environment, understanding how formation of domain walls in superconductor will modify the Ginzburg-Landau free energy of the superconductor. Then we will put the superconductor in a static magnetic field, and explore how that will give rise to the difference in type-I and type-II superconductor.

In Ginzburg-Landau theory for superconductor, a complex-valued order parameter Ψ is introduced, such that $\Psi = 0$ in the normal state and $\Psi \neq 0$ in the superconducting state of the material. Suppose that at some critical temperature T_c, the material changes from its normal state to superconducting state. Let us start with the Ginzberg-Landau free energy density in the absence of magnetic fields, [1]

$$f = \frac{\hbar^2}{4m}|\nabla\Psi|^2 + \alpha t|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 \,,$$

where α, β are constants and $t = (T - T_c)/T_c$. We see that for $t < 0$, the free energy is minimized when

$$\Psi_* = \sqrt{-\frac{\alpha t}{\beta}} e^{i\phi}$$

with ϕ being the phase of the superconductor, taken to be a constant. Below the critical temperature, we see that the superconducting state has a lower free energy than the normal state,

$$f(\Psi_*) < f(\Psi = 0) \,,$$

and thus is thermodynamically favorable. Such an energy difference can be understood via the Meissner effect in superconductivity, where we need to define a temperature-dependent ($T < T_c$) critical magnetic field H_c such that

$$f(\Psi = 0) - f(\Psi_*) = \frac{H_c^2}{8\pi} \,, \qquad \text{thus } \frac{(\alpha t)^2}{\beta} = \frac{H_c^2}{4\pi} \,.$$

The magnetic language enters when comparing superconducting and normal states in an applied field. Physically, superconductivity repels external magnetic field below H_c, known as the Meissner effect. $H_c^2/(8\pi)$ itself can be understood as the energy stored in a magnetic field H_c, but here it serves as a measure of the amount of electron pairing and condensation energy stored in the superconducting state of the material.

Domain Wall in Superconductor

Here we begin our study of domain wall in superconductor, the interface surface separating the superconducting state and the normal state of the material, in Ginzburg-Landau theory. Let us denote $K \coloneqq \hbar^2/(4m)$. A length scale can be obtained from here,

$$\xi \coloneqq \left(\frac{K}{|\alpha t|}\right)^{1/2} = \frac{\hbar}{2\sqrt{m\alpha|t|}} \,,$$

[1] This equation corresponds to Eq. (5.46) from [1], and Eq. (1.38) from [2].

which is called the coherence length. Varying the Landau free energy $F = \int d^3x\, f$ with respect to Ψ^* one obtains

$$\frac{\delta F}{\delta(\Psi^*)} = \alpha t \Psi + \beta |\psi|^2 \Psi - K \nabla^2 \Psi \,,$$

We can rescale the equation with respect to Ψ_*,

$$\Psi = \sqrt{\frac{|\alpha t|}{\beta}}\, \psi$$

such that, when setting the variation to be zero for minimizing F,

$$-\xi^2 \nabla^2 \psi + \mathrm{sgn}(t)\, \psi + |\psi|^2 \psi = 0 \,.$$

From here we look for a domain wall solution: Consider $T < T_c$, and assume that all spatial variation occurs in the x-direction, with boundary condition $\psi(x = 0) = 0$ and $\psi(x = \infty) = 1$. Assuming $\psi(x) = f(x)\, e^{i\alpha}$ with α being a constant, we then have

$$-\xi^2 f'' - f + f^3 = 0 \,,$$

which can be solved via

$$f(x) = \tanh\left(\frac{x}{\sqrt{2}\xi}\right) \,,$$

from which we see that the width of such a domain wall is of order ξ. We now want to ask whether the formation of such a domain wall is energetically favorable in superconductor. We can first calculate the energy per unit area stored in such a domain wall solution f, via integrating over x,

$$\bar{\sigma} = \int_0^\infty dx \left(K|\partial_x \Psi|^2 + \alpha t |\Psi|^2 + \frac{\beta}{2}|\Psi|^4 \right) = \frac{(\alpha t)^2}{b} \int_0^\infty dx \left(\xi^2 (\partial_x f)^2 - f^2 + \frac{f^4}{2} \right) .$$

Then the energy difference σ, the true domain wall energy relative to the superconducting state, is

$$\sigma = \bar{\sigma} - \int_0^\infty dx \left(-\frac{H_c^2}{8\pi} \right) = \frac{(\alpha t)^2}{\beta} \int_0^\infty dx \left(\xi^2 (\partial_x f)^2 + \frac{(1 - f^2)^2}{2} \right) := \frac{H_c^2}{8\pi} \delta \,,$$

where

$$\delta = 2 \int_0^\infty dx\, (1 - f^2)^2 = \frac{4\sqrt{2}}{3} \xi \,.$$

This shows that σ is positive, and thus the domain wall is energetic unfavorable, the positive surface energy keeps the system spatially homogeneous for all $H < H_c$. However, for superconductor placed in a magnetic field, the field lines can in fact penetrate into the superconductor by a penetration depth λ_L.[2] If such an effect is included in the analysis, one can show that $\delta < 0$ for $\lambda_L \gg \xi$ and thus domain wall is energetic favorable in that case, the negative surface energy causes the system to break into domains as soon as soon as H exceeds some critical field H_{c1}. We will demonstrate this effect by considering placing the superconductor in a static magnetic field, in the following discussion.

[2] λ_L is called δ in Section 5.7 of [1], the London penetration depth of the magnetic field into a superconductor.

Superconductor in a Static Magnetic Field

We start with the Ginzberg-Landau free energy functional in the presence of a static vector potential $\vec{A}(\vec{x})$, [3]

$$F = F_n + \int d^3x \left(\frac{(\nabla \times \vec{A})^2}{8\pi} + \frac{\hbar^2}{4m} \left| \left(\nabla - \frac{2ie}{\hbar c}\vec{A} \right) \Psi \right|^2 + \alpha t |\Psi|^2 + \frac{\beta}{2}|\Psi|^4 \right) ,$$

where F_n is the normal-state free energy. The first term in the integrant is obtained from the energy density of the bare magnetic field, and the second term is obtained from coupling the complex-valued order parameter ψ to the magnetic vector potential $\vec{A}$. The differential free energy density of a magnetic material is given by [4]

$$df = -s\, dT + \frac{1}{4\pi}\vec{H} \cdot d\vec{B} \,.$$

For describing superconductor in applied field, we shall switch to the Gibbs free energy framework because the natural control variable in the experiment is the applied field $\vec{H}$ rather than the internal induction $\vec{B}$. Once the sample magnetizes, $\vec{B}$ is not directly under control. Therefore, we derive

$$G = F_n + \int d^3x \left(\frac{B^2}{8\pi} - \frac{\vec{H} \cdot \vec{B}}{4\pi} + \frac{\hbar^2}{4m} \left| \left(\nabla - \frac{2ie}{\hbar c}\vec{A} \right) \Psi \right|^2 + \alpha t |\Psi|^2 + \frac{\beta}{2}|\Psi|^4 \right)$$

Now we consider an applied field $H = H_c$, with coexistence of normal and superconducting states in the material. The formation of planer interface in the material between the two states requires energy, the Gibbs free energy of the domain wall. Suppose that we have superconducting state on the right ($\psi(x) = 1$ at $x \to \infty$) and the normal state on the left ($\psi(x) = 0$ at $x \to -\infty$) Technically, we shall solve for Ψ by minimizing G under these boundary conditions about Ψ, which can be done numerically [3]. Here we will only derive the excess energy due to the presence of the domain wall, by first noticing that the Gibbs free energy densities at the infinities are equal, that is

$$g_s = g(x \to \infty) = g(x \to -\infty) = g_n \,.$$

This is because, for the normal state, $\Psi = 0$ and $H_c = B$,

$$g(x \to -\infty) = f_n(x \to -\infty) + \frac{H_c^2}{8\pi} - \frac{H_c^2}{4\pi} = f_n - \frac{H_c^2}{8\pi} \,;$$

For the superconducting state, $B = 0$ and $A = 0$ with a fixed gauge, thus

$$g(x \to \infty) = f_n(x \to \infty) + \frac{\hbar^2}{4m} \left| \left(\nabla - \frac{2ie}{\hbar c}\vec{A} \right) \Psi \right|^2 + \alpha t |\Psi|^2 + \frac{\beta}{2}|\Psi|^4 = f_n - \frac{H_c^2}{8\pi} \,.$$

We conclude that the excess Gibbs free energy density due to the domain wall is

$$\Delta g = g(x) - g_s = \frac{B^2}{8\pi} - \frac{\vec{H}_c \cdot \vec{B}}{4\pi} + \frac{\hbar^2}{4m} \left| \left(\nabla - \frac{2ie}{\hbar c}\vec{A} \right) \Psi \right|^2 + \alpha t |\Psi|^2 + \frac{\beta}{2}|\Psi|^4 - \frac{H_c^2}{8\pi} \,.$$

[3] This equation corresponds to Eq. (5.56) [1], and Eq. (1.47) from [2].

[4] This is Eq. (1.2) in [2]

To simplify our calculation, we define $b = B/(\sqrt{2}H_c)$, $h = H/(\sqrt{2}H_c)$, $\vec{A} = \sqrt{2}\lambda_L H_c \vec{a}$, and use the London penetration depth [5]

$$\lambda_L^{-2} = 8\pi K \left(\frac{2e}{\hbar c}\right)^2 |\Psi|^2 ,$$

such that

$$\Delta G = \frac{H_c^2 \lambda_L^3}{4\pi} \int d^3r \left(-|\psi|^2 + \frac{|\psi|^4}{2} + |(\kappa^{-1}\nabla + i\vec{a})\psi|^2 + (b-h)^2 \right)$$

with $\Psi = \sqrt{|\alpha t|/\beta}\psi$, $\vec{x} = \lambda_L \vec{r}$, and the Ginzberg-Landau parameter

$$\kappa := \frac{\lambda_L}{\xi} = \frac{2\sqrt{2}e}{\hbar c} H_c \lambda_L^2 .$$

We have picked $H = H_c$, thus $h = 1/\sqrt{2}$. We can also fix the gauge of $\vec{A}$ to study the case where

$$\vec{A}(x) = A_y(x)\,\hat{y} , \qquad \vec{B}(x) = B_z(x)\,\hat{z} , \qquad b = a'$$

After these simplifications, we are left to study the minimization of the functional

$$I_\kappa[f,a] = \int_{-\infty}^{\infty} dr \left(-f^2 + \frac{f^4}{2} + \frac{(f')^2}{\kappa^2} + a^2 f^2 + (a'-h)^2 \right) , \qquad \text{with } h = \frac{1}{\sqrt{2}} , \qquad (*)$$

subject to boundary conditions for $b = a'$ and real-valued $f = \psi$,

$$f(-\infty) = 0 , \qquad f(\infty) = 1 , \qquad b(-\infty) = h , \qquad b(\infty) = 0 .$$

The functional I_κ dictates the sign of the domain wall energy per unit area

$$\sigma = \frac{H_c^2 \lambda_L}{4\pi} I_{\kappa,\min} .$$

Varying I_κ with respect to f and a for minimization of I_κ gives the coupled Ginzburg-Landau equations for the domain wall,

$$\frac{1}{\kappa^2} f'' - (a^2-1)f - f^3 = 0 , \qquad a'' = f^2 a . \qquad (\dagger)$$

Notice that

$$\frac{d}{dr}(ff') = (f')^2 + ff'' .$$

Since $f' = 0$ at the boundary $r = \pm\infty$, we can write

$$\int_{-\infty}^{\infty} dr \left(-f^2 + f^4 + a^2 f^2 + \frac{(f')^2}{\kappa^2} \right) = 0 .$$

Combining with (*), we see that the minimization of I_κ occurs at

$$I_{\kappa,\min} = \int_{-\infty}^{\infty} \left(\left(b - \frac{1}{\sqrt{2}} \right)^2 - \frac{1}{2} f^4 \right) , \qquad \text{when } a'' = f^2 a . \qquad (**)$$

The physical meaning of (**) is now clear. The domain wall energy is given by the difference of the energy cost of expelling the magnetic field (the first term) and the energy gain due to superconducting condensation.

[5] This is introduced in section 5.7 of [1] and is derived in Section 1.4.3 of [2]. Notice that $|\Psi|^2 = |\alpha t|/\beta$ in the superconducting state.

Type-I/II Superconductor and Ginzburg-Landau Parameter

Next, we minimizes (*) with $\kappa = 1/\sqrt{2}$, which we will soon see that it signifies the difference between type-I and type-II superconductor. From observation, we have

$$\begin{aligned} I_{\kappa=1/\sqrt{2}} &= \int_{-\infty}^{\infty} dr \left(-f^2 + \frac{f^4}{2} + 2(f')^2 + a^2 f^2 + \left(a' - \frac{1}{\sqrt{2}} \right)^2 \right) \\ &= \int_{-\infty}^{\infty} dr \left((\sqrt{2} f' + af)^2 + \left(b - \frac{1-f^2}{\sqrt{2}} \right)^2 - \frac{d}{dr}(\sqrt{2} a f^2) \right) \\ &= \int_{-\infty}^{\infty} dr \left((\sqrt{2} f' + af)^2 + \left(b - \frac{1-f^2}{\sqrt{2}} \right)^2 \right) , \end{aligned}$$

which is a sum of squares, thus $\sigma \propto I_{\kappa=1/\sqrt{2}} \geq 0$. In particular, the minimization $I_{\kappa=1/\sqrt{2}} = 0$ occurs when we have

$$\sqrt{2} f' + af = 0 \,, \qquad \text{and } b = a' = \frac{1-f^2}{\sqrt{2}} \,. \tag{‡}$$

Indeed, they together solves the Ginzburg-Landau equations (†) for the domain wall, this is seen by first differentiating the first equation in (‡)

$$\frac{d}{dr}\left(\sqrt{2} f' + af \right) = \sqrt{2} f'' + a' f + a f' \,,$$

then combine with the second equation in (‡),

$$2f'' + (1-f^2) f - a^2 f = 0$$

which is exactly the first equation in (†) when $\kappa = 1/\sqrt{2}$. Likewise, differentiating the second equation in (‡) gives

$$a'' = b' = -\sqrt{2} f f' \,,$$

and combining with the first equation in (‡) we obtain

$$a'' = f^2 a \,,$$

which is exactly the second equation in (†). This shows that, when $\kappa = 1/\sqrt{2}$, the minimization of I_κ via f and a gives exactly $I_{\kappa,\min} = 0$, thus the energy cost of creating a domain wall is $\sigma = 0$.

Our final task is to demonstrate that the minimized $I_{\kappa,\min}$ is monotone decreasing across $\kappa = 1/\sqrt{2}$. With this established, we will have shown that, as $\sigma \propto I_{\kappa,\min}$, the domain wall is energetic unfavorable when $\kappa < 1/\sqrt{2}$ and favorable when $\kappa > 1/\sqrt{2}$. This is straightforward, for f_1, a_1 minimizing I_{κ_1} and f_2, a_2 minimizing I_{κ_2}, with $0 < \kappa_1 < \kappa_2$,

we see that

$$
\begin{aligned}
I_{\kappa_2,\min} = I_{\kappa_2}[f_2, a_2] &\leq I_{\kappa_2}[f_1, a_1] \\
&= \int du \left(-f_1^2 + \frac{f_1^4}{2} + \frac{(f_1')^2}{\kappa_2^2} + a_1^2 f_1^2 + (a_1' - h)^2 \right) \\
&= \int du \left(-f_1^2 + \frac{f_1^4}{2} + \frac{(f_1')^2}{\kappa_2^2} + a_1^2 f_1^2 + (a_1' - h)^2 + \frac{(f_1')^2}{\kappa_1^2} - \frac{(f_1')^2}{\kappa_1^2} \right) \\
&= \int du \left(-f_1^2 + \frac{f_1^4}{2} + \frac{(f_1')^2}{\kappa_1^2} + a_1^2 f_1^2 + (a_1' - h)^2 + \left(\frac{1}{\kappa_2^2} - \frac{1}{\kappa_1^2} \right) (f_1')^2 \right) \\
&< \int du \left(-f_1^2 + \frac{f_1^4}{2} + \frac{(f_1')^2}{\kappa_1^2} + a_1^2 f_1^2 + (a_1' - h)^2 \right) \\
&= I_{\kappa_1}[f_1, a_1] \leq I_{\kappa_2}[f_1, a_1] = I_{\kappa_1,\min} \,.
\end{aligned}
$$

for f_1' be nonzero in a non-measure-zero set, which is required for having a domain wall in space. We now concludes that $I_{\kappa,\min}$ flips sign across $\kappa = 1/\sqrt{2}$, and thus the critical value $\kappa = 1/\sqrt{2}$ distinguishes type-I and type-II superconductors.

In type-I superconductor, $\kappa < 1/\sqrt{2}$, where we have seen that the energy cost for formation of the domain wall σ is positive in this region of κ, thus preventing the system breaking into domains. In contrast, for type-II superconductor, $\kappa > 1/\sqrt{2}$, and the negativity in σ favors the formation of domain walls, leading to a spontaneous formation of domains containing vortices that carry magnetic flux.

References

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[2] Daniel Arovas and Congjun Wu, *Lecture Notes on Superconductivity*, University of California, San Diego (2019).

[3] Carsten Timm, *Theory of Superconductivity*, Technische Universität Dresden, (2026).